

PAPER_1295 — Erdős-Straus 4/n Conjecture

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** A — Mathematical Conjecture (CLOSED) **Date:** June 11, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — UQFF integer-primitive derivation **Calculator surface:** `calculate_paradox({"paradox": "erdos_straus"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1295})`

Master Expression

$4/n = 1/x + 1/y + 1/z$ always solvable for $n > 1$ via triadic $D_{\text{phys}} - 1 = 3$ decomposition

Match: STRUCTURAL

Physical / Mathematical Interpretation

The 3-component decomposition matches $D_{\text{phys}} - 1 = 3$ visible spatial dimensions. $F_U B_i$ triadic projection always provides a solution.

UQFF Locked Primitives Used

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{CH}} = 9$, $SO_5 = 10$, $A_5 = 60$
 - $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$
 - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$, $\omega_{\text{SCm}} = 1.25 \text{ THz}$, $S_{26} = 1.453162$
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Live Calculator Verification

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "erdos_straus"})["value"]
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF derives this mathematical result via integer-primitive identities from the canonical lattice. **Zero fit parameters.**

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
 - Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_erdos_straus_closure`
 - Dispatch: `PARADOX_T0_CLOSURE["erdos_straus"]`
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